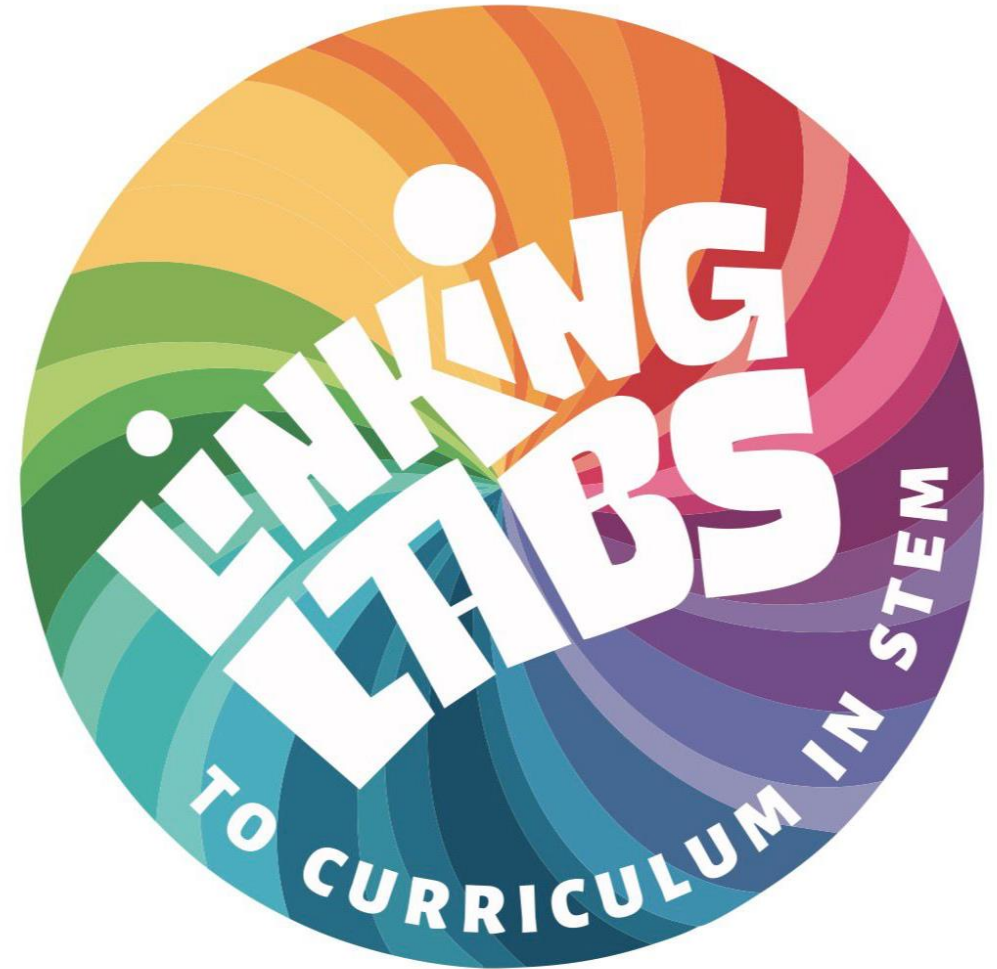


Today's agenda

- First activity: Predictions of last week for our Lemna experiment + data collection
- Second activity: Seeing Microorganisms living in Pond Water + Reviewing how to use a microscope
- Third activity: What is on the menu? Looking at slides under the microscope





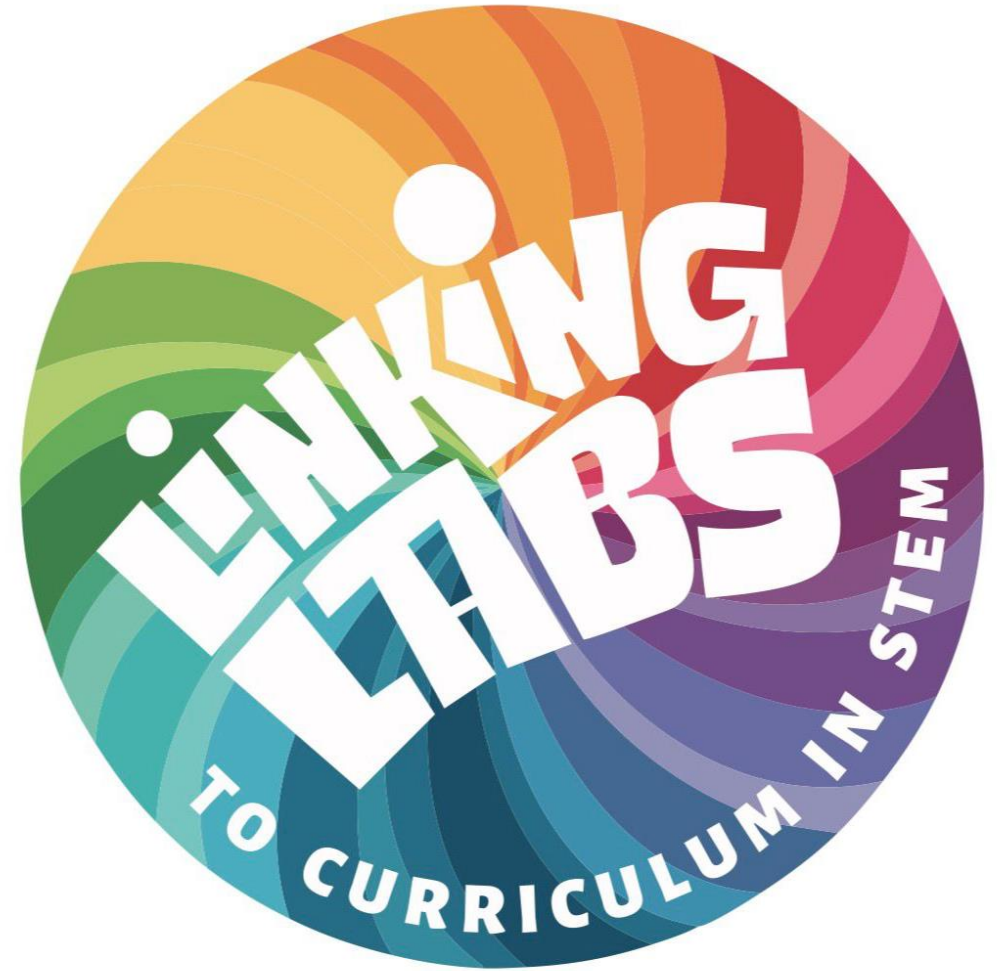
Proper lab attire

- ✓ Tie long hair back
- ✓ Button-up lab coat all the way
- ✓ Wear long pants or long skirt
- ✓ Closed-toed shoes



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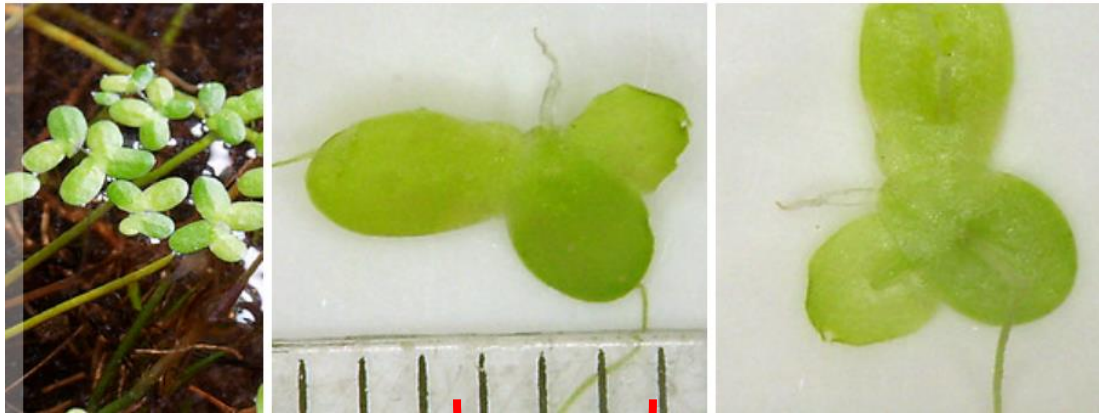


Which were the predictions?

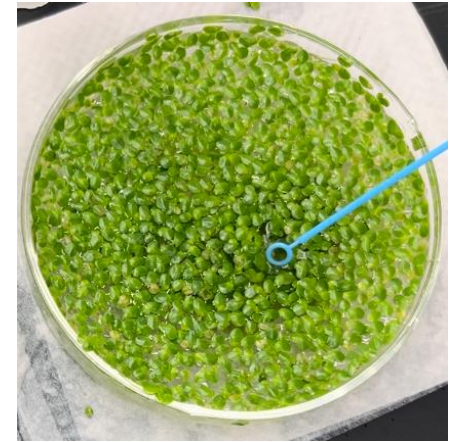
Does different types of salt affect the environment?

Lemna minor

Set up experiment



One thalli



Does different types of salt affect the environment?

Set up experiment



**TAKE NOTES OF HOW
MANY PLANTS AND LEAFS
YOU HAVE IN EACH CUP**



Do different types of salt affect the environment?

Set up experiment



**TAKE NOTES OF HOW
MANY PLANTS AND LEAFS
YOU HAVE IN EACH CUP**

**Collect the results in
One week**

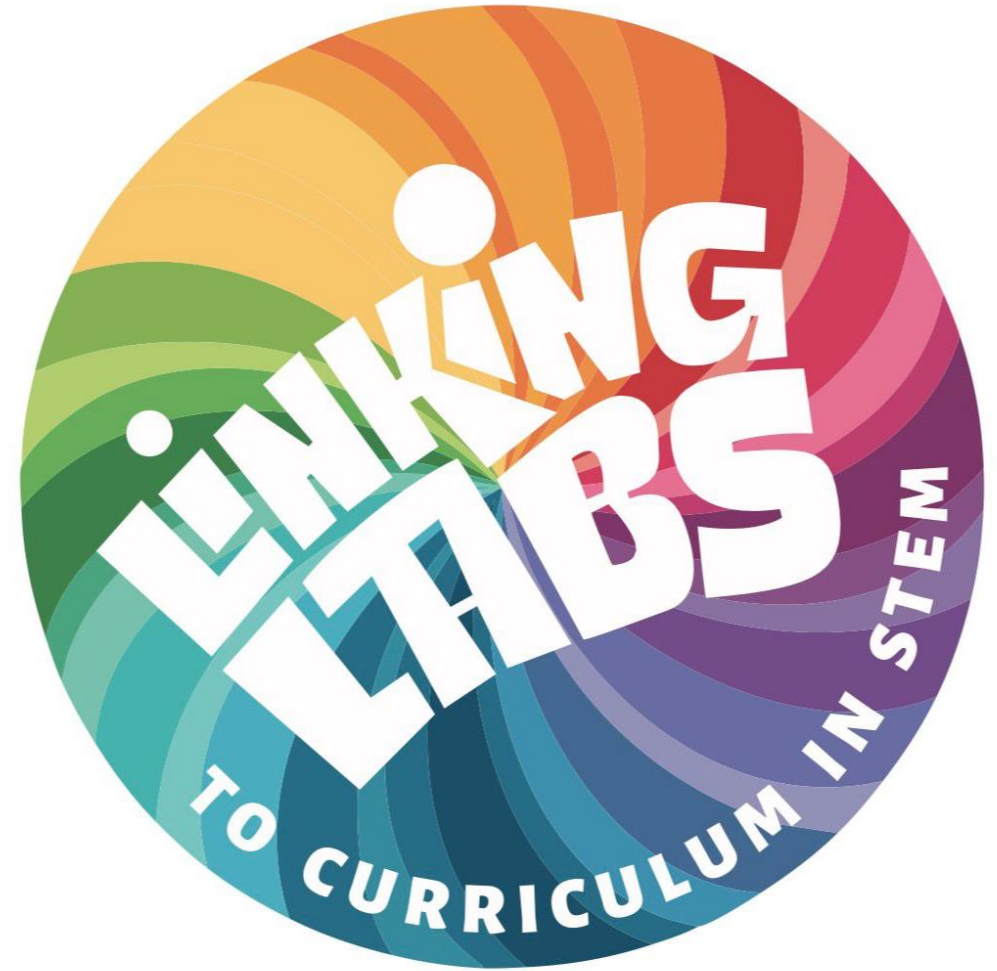


Does different types of salt affect the environment?

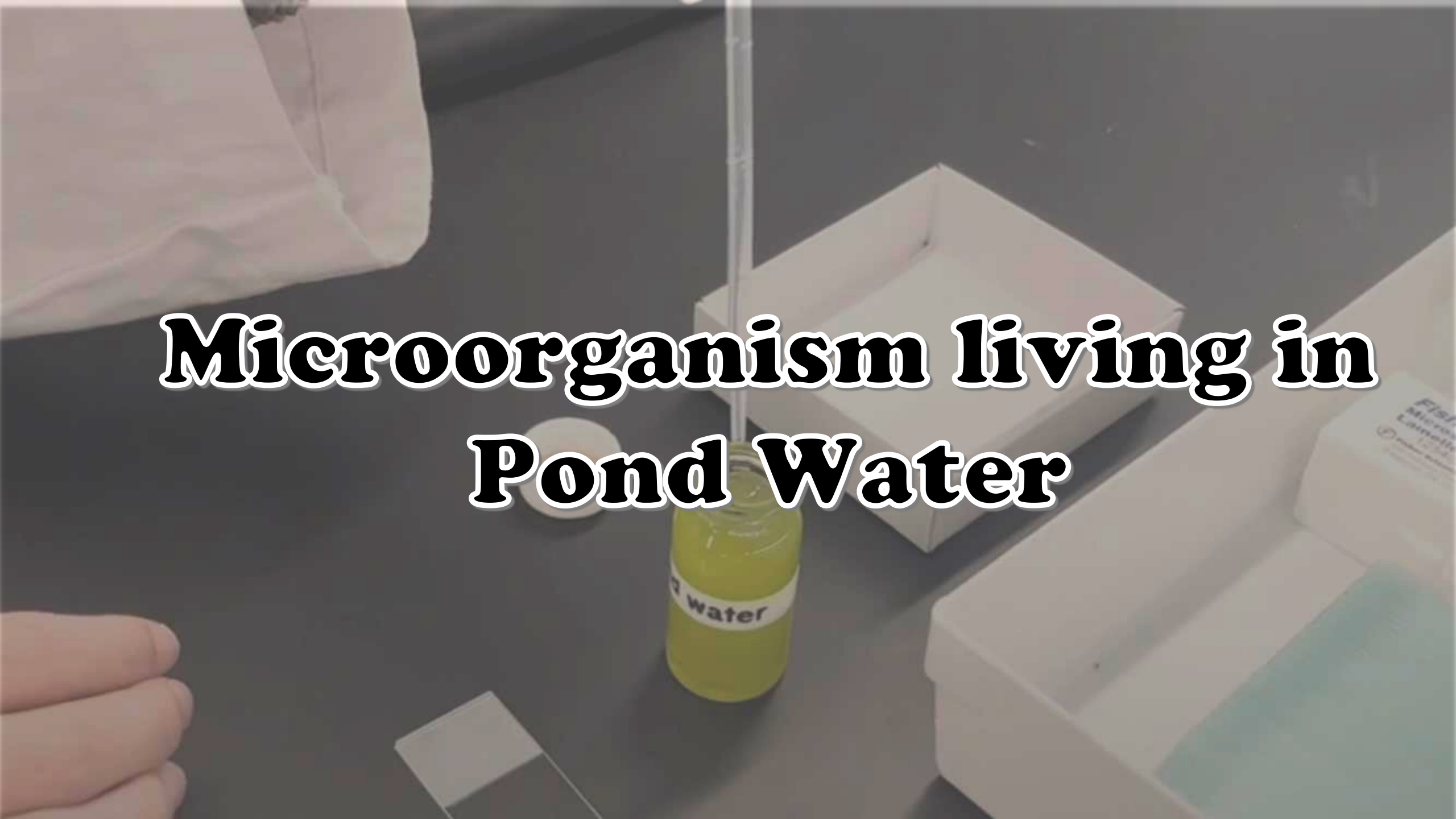
Conditions	Initial number of plants	Final number of plants	Observation and Descriptions
Positive Control (No added salts)			
Negative Control (20ml mixed road salts)			
NaCl Solution (20ml sodium chloride)			
KCl Solution (20ml potassium chloride)			
CaCl ₂ Solution (20ml calcium chloride)			

Today's agenda

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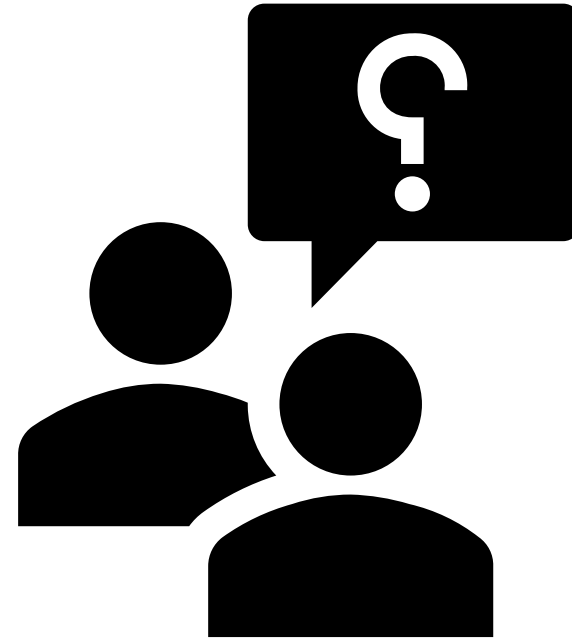
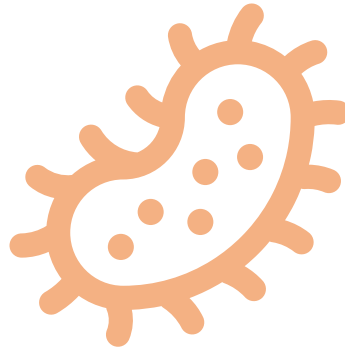


Microorganism living in Pond Water



Microorganisms

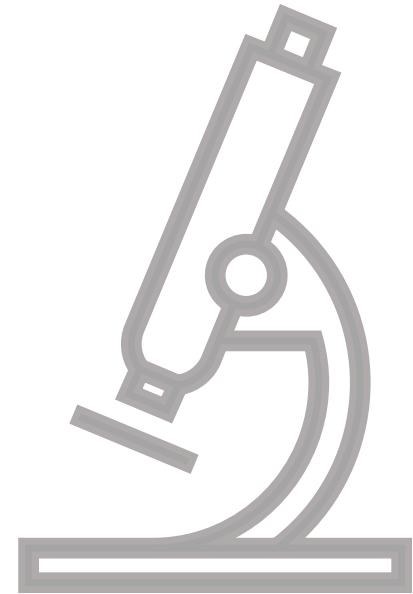
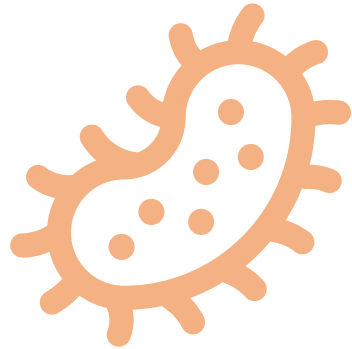
What are microorganisms?



Microorganisms

What are microorganisms?

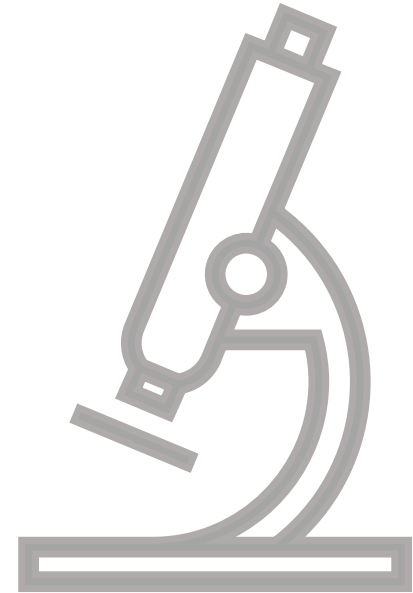
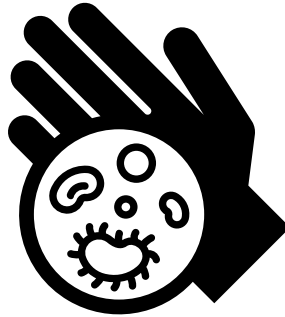
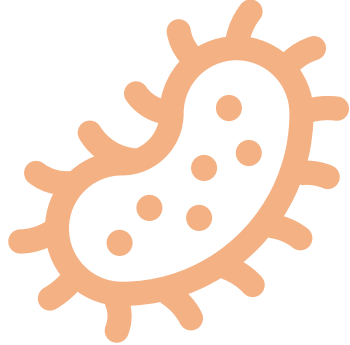
Organisms that cannot be seen by the naked eye and require a microscope to be observed



Microorganisms

What are microorganisms?

Organisms that cannot be seen by the naked eye and require a microscope to be observed



They are classified differently based on their cellular structure and way to obtain energy

Cellular structure

Eukaryotes vs. Prokaryotes

Cellular structure

Eukaryotes vs. Prokaryotes



Eukaryotes are species with a cell that has a nucleus



Prokaryotes are species with a cell that does not have a nucleus

Unicellular vs. Multicellular

Unicellular vs. Multicellular



Unicellular organisms are made up of one cell. e.x. Unicellular algae



Multicellular organisms are made up of many cells. e.x. Seaweed

How they obtain energy

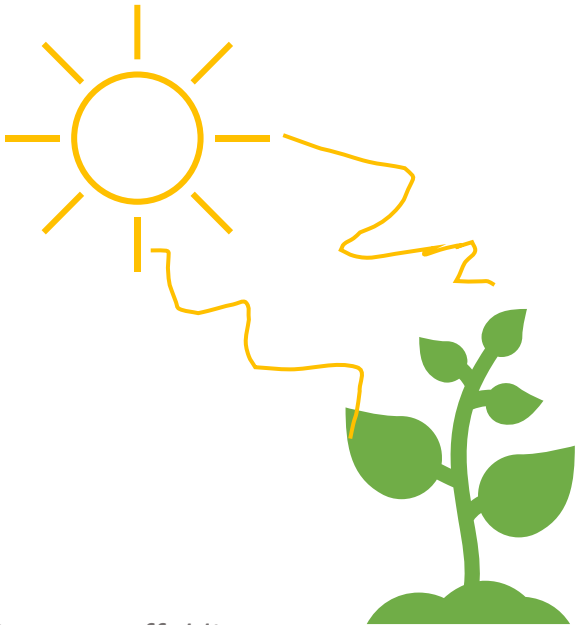


How they obtain energy



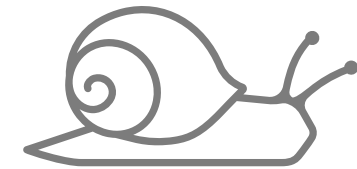
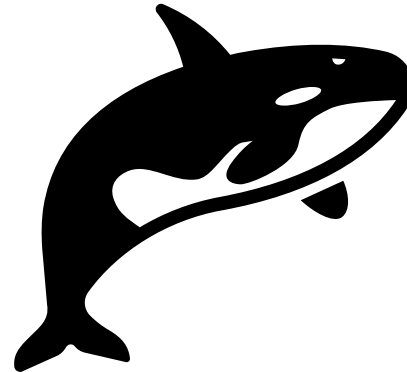
Photoautotrophs

Plants and other photosynthetic organisms capture sunlight energy to covert carbon dioxide into sugars.



Chemoheterotrophs

Animals and organisms that consume other organisms to obtain energy and carbon.



Microorganisms living in Pond Water



Microorganisms living in Pond Water



Ponds have a high richness of aquatic plant, micro and macroinvertebrate

What type of microorganism live in pond water?

What type of microorganism live in pond water?

Green Algae

Protozoa

Rotifers

What type of microorganism live in pond water?

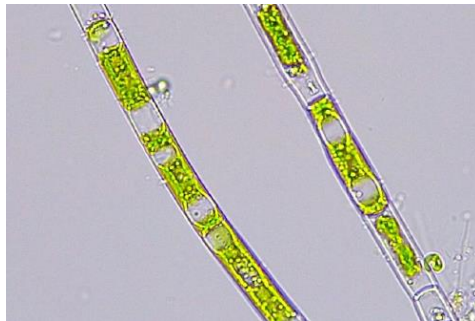
Green Algae

Protozoa

Rotifers

Mobility

Energy



What type of microorganism live in pond water?



Cyanobacteria

© Dr. Robert Calentine/Visuals Unlimited



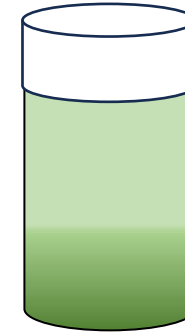
Bacteria

© Bryan/Sui Generis Brewing



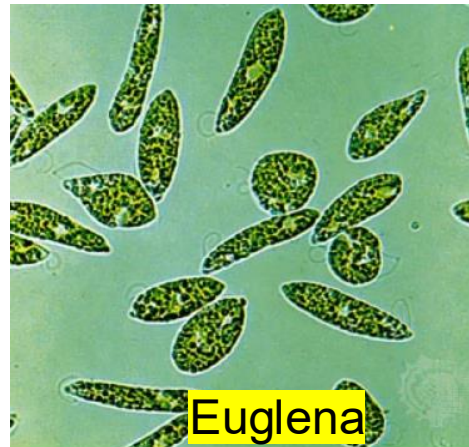
Ciliates

© Microscopy UK/Wim Van Egmond



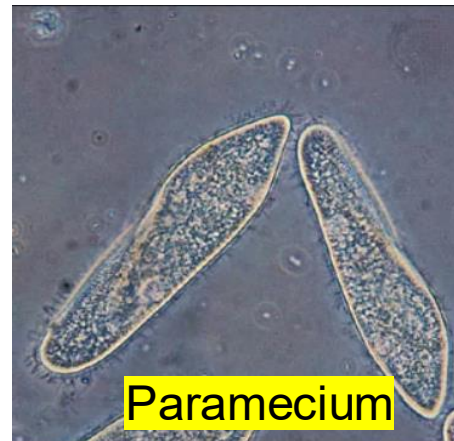
Amoeba

© Russ Kinne/Photo Researchers



Euglena

© Walter Dawn



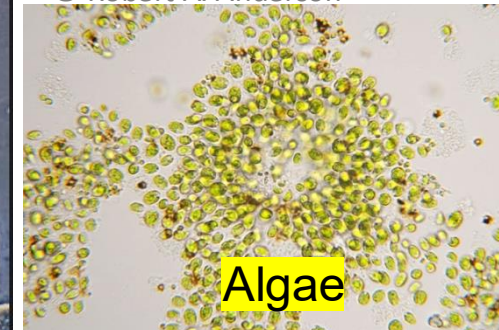
Paramecium

© John J. Lee



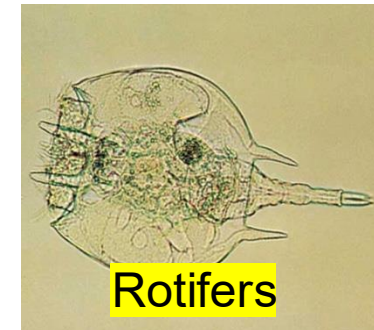
Diatoms

© Robert A. Anderson



Algae

© NNehring/iStock.com



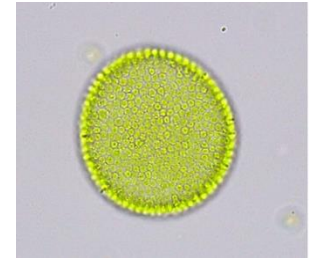
Rotifers

© Runk/Schoenberger

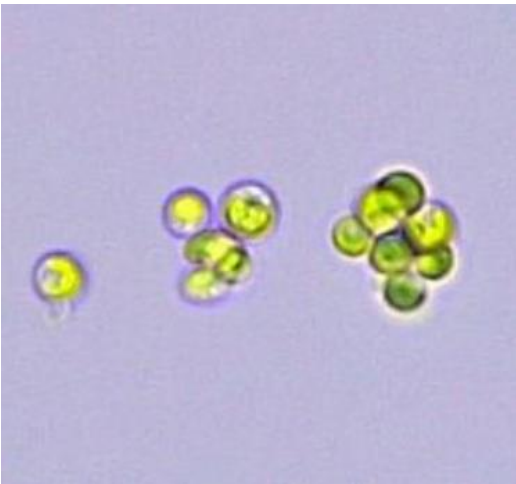
Identifying Different Organisms

- Green Algae

Volvox



Chlorella



Scenedesmus



Selenastrum



Ulothrix

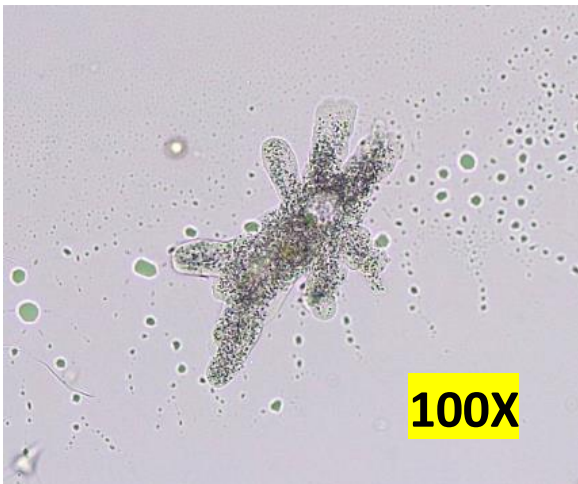


ALL PICTURES TAKEN IN 400X MAGNIFICATION

Identifying Different Organisms

- Protozoa

Amoeba



Euglena gracilis



Paramecium aurelia



Identifying Different Organisms

- Rotifers

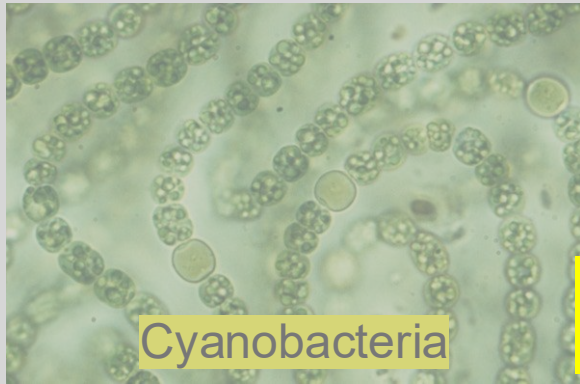
Monostyla clostercerca



Philodina acuticornis



What type of microorganism live in pond water?

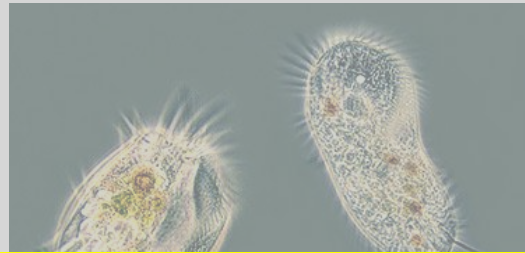


Cyanobacteria

© Dr. Robert Calentine/Visuals Unlimited

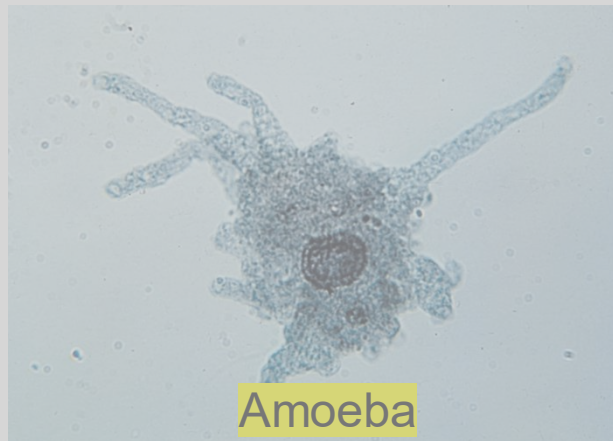


© Bryan/Sui Generis Brewing



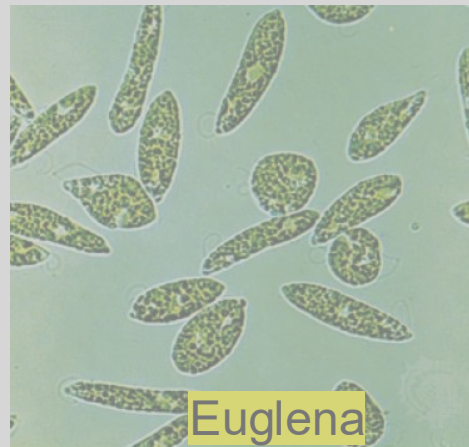
© Microscopy UK/Wim Van Egmond

HOW CAN WE OBSERVE THEM?



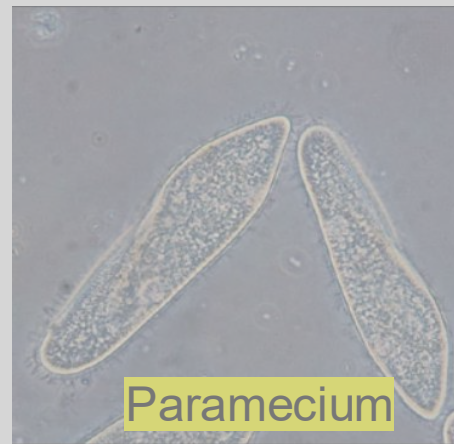
Amoeba

© Russ Kinne/Photo Researchers



Euglena

© Walter Dawn



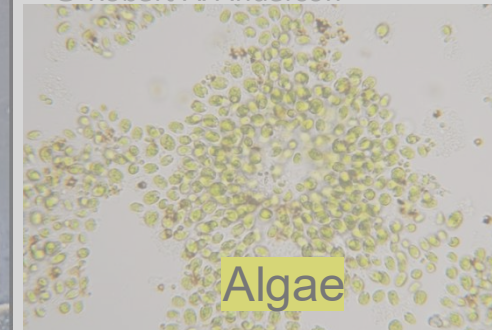
Paramecium

© John J. Lee



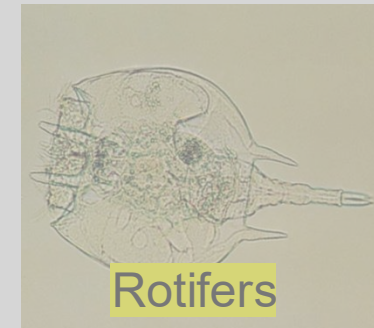
Diatoms

© Robert A. Anderson



Algae

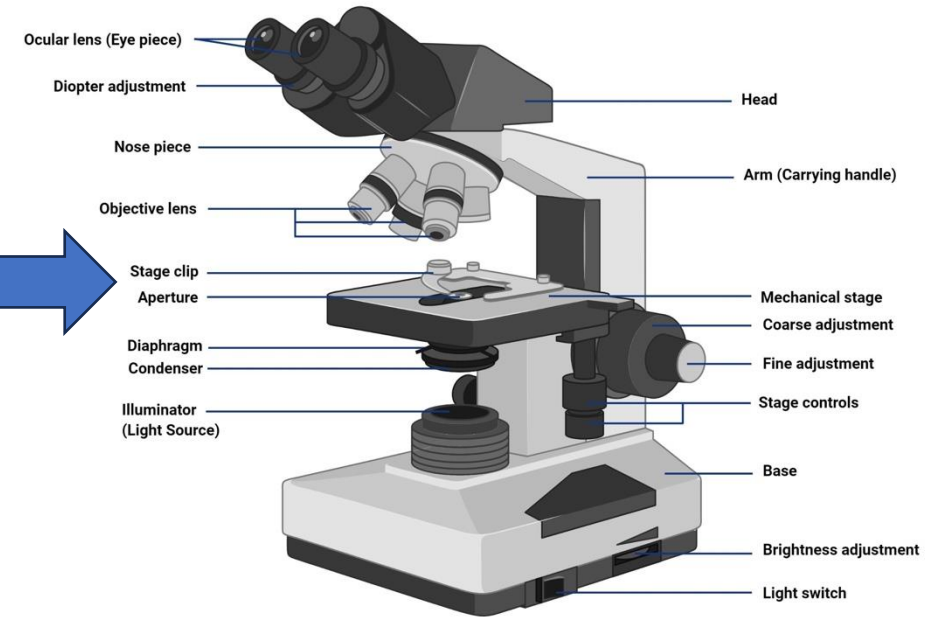
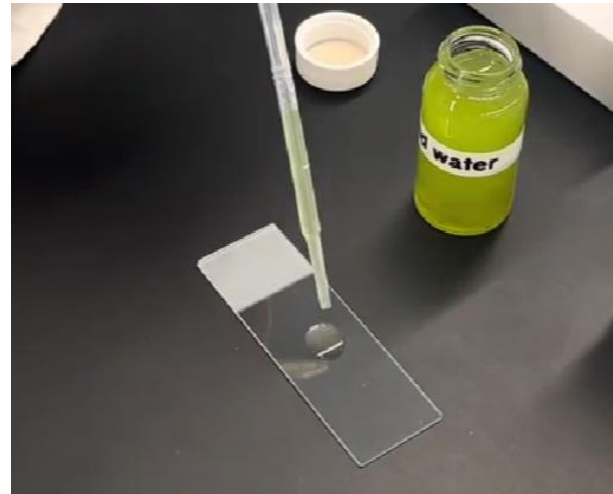
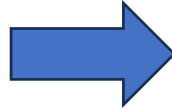
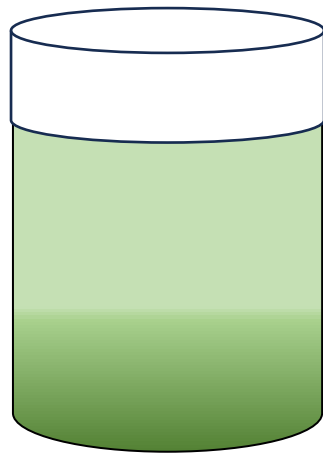
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Rotifers

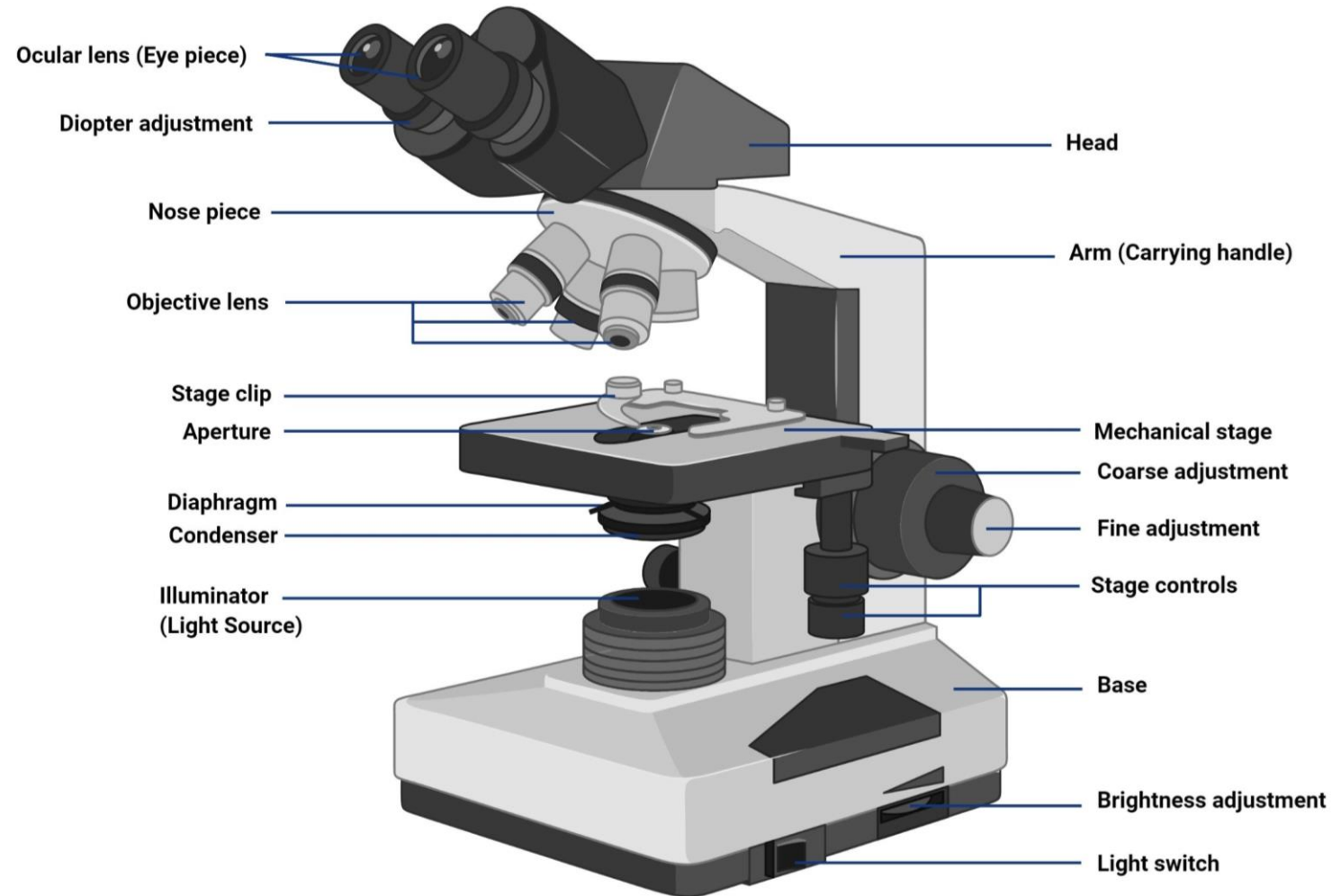
© Runk/Schoenberger

HOW CAN WE OBSERVE THEM?



© Sagar Aryal

Compound Microscope



© Sagar Aryal

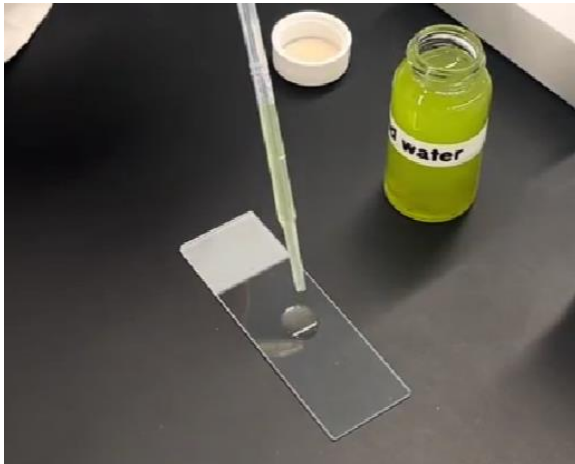
Compound Microscope



Collecting observation data

Working in groups

1. Shake up container of recently lab made pond water
2. Pipette 1 drop onto a cover slide
3. Place coverslip onto the drop at a 45-degree angle
4. Take notes of the different microorganism that you observe



SCAVENGER HUNT GAME: Point System



Volvox : 2 POINTS



Rotifers: 4 & 3 points



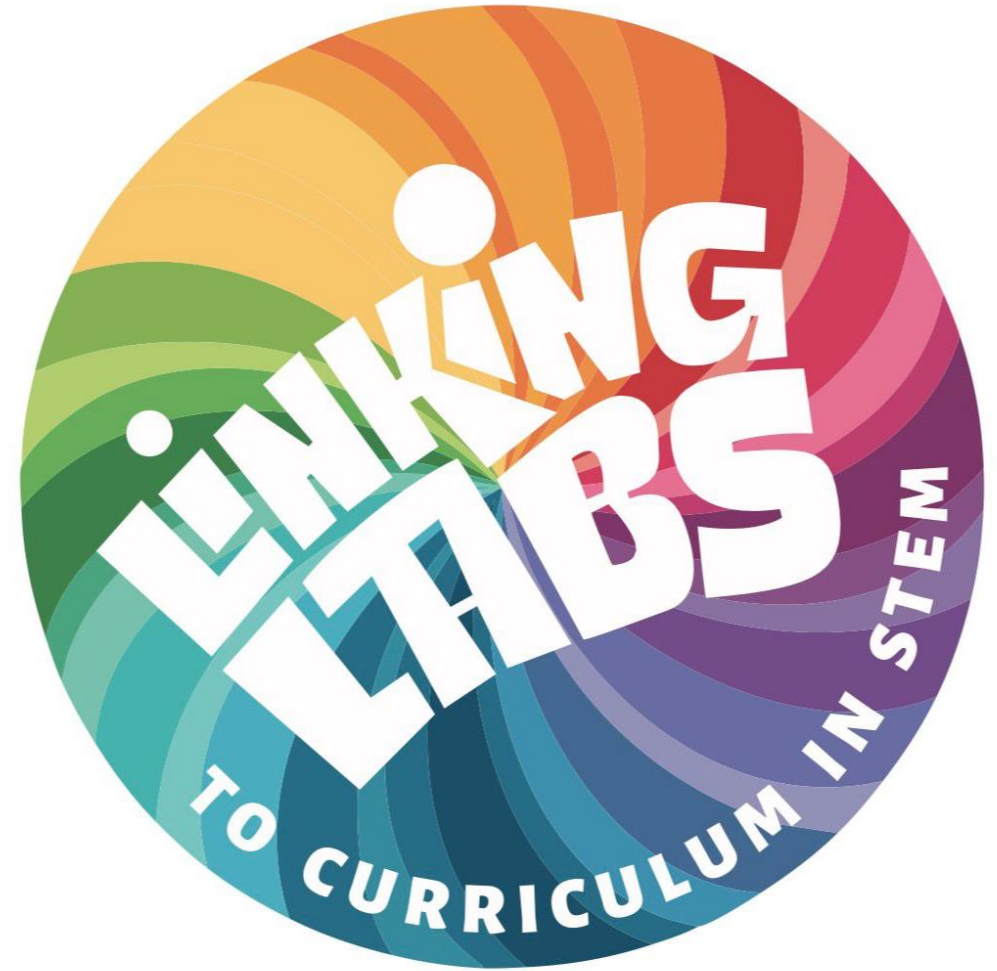
(CALL US TO CONFIRM)

CLEAN UP

- Discard the slides and pipets in the correct containers

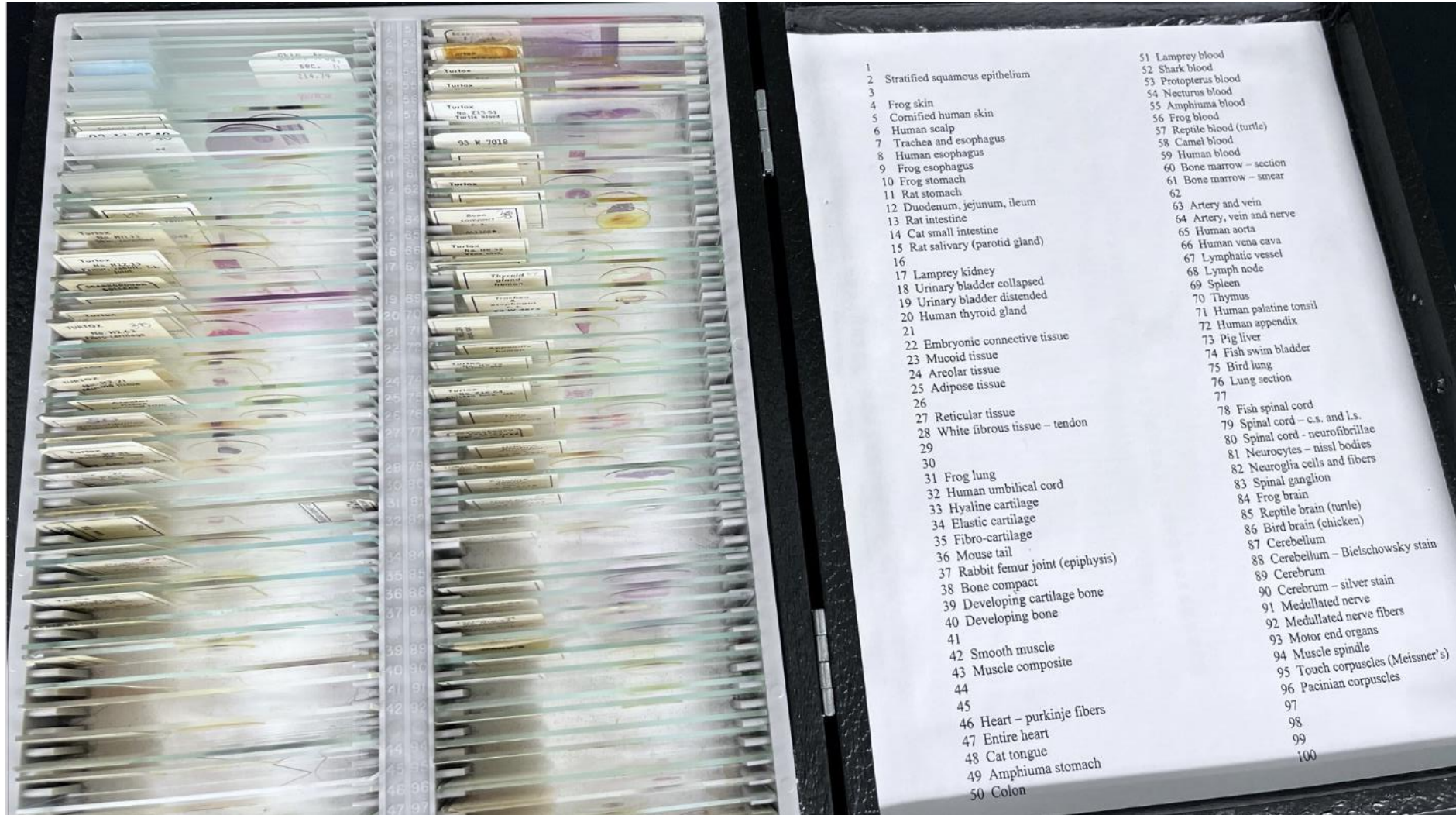
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What is on the menu?

Looking at slides under the microscope



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<https://linc-stem-program.github.io/>

<https://www.utsc.utoronto.ca/labs/andrade/linc-stem/>

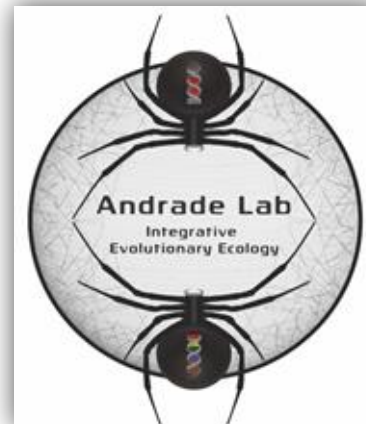
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<https://github.com/Labs-LinC-STEM/Breaking-Barriers-to-Science>

Support

Financially:

In-kind support:

